

Double-click (or enter) to edit

```
import os
os.listdir()
```

```
['.config', 'sample_data']
```

```
!pip install -q pyspark
```

```
_____ 45.9/45.9 kB 2.4 MB/s eta 0:00:00
_____ 398.7/398.7 kB 10.7 MB/s eta 0:00:00
_____ 12.5/12.5 MB 85.0 MB/s eta 0:00:00
_____ 4.9/4.9 MB 94.5 MB/s eta 0:00:00
_____ 77.5/77.5 kB 4.8 MB/s eta 0:00:00
_____ 59.8/59.8 kB 3.6 MB/s eta 0:00:00
```

```
from pyspark.sql import SparkSession

spark=SparkSession.builder \
.master("local[*]") \
.appName("colabSpark")\
.getOrCreate()
print("spark version",spark.version)
```

```
spark version 4.0.2
```

Question 1)

```
df = spark.read.csv("heroes_information.csv", header=True)
df.show() # by default 20 observation
```

Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publisher	Skincolor	Alignment	Weight
0	A-Bomb	Male	yellow	Human	No Hair	203	Marvel Comics	-	good	441
1	Abe Sapien	Male	blue	Ichthyo Sapien	No Hair	191	Dark Horse Comics	blue	good	65
2	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
3	Abomination	Male	green	Human / Radiation	No Hair	203	Marvel Comics	-	bad	441
4	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	-99
5	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
6	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	-99
7	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
8	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
9	Agent Bob	Male	brown	Human	Brown	178	Marvel Comics	-	good	81
10	Agent Zero	Male	-	-	-	191	Marvel Comics	-	good	104
11	Air-Walker	Male	blue	-	White	188	Marvel Comics	-	bad	108
12	Ajax	Male	brown	Cyborg	Black	193	Marvel Comics	-	bad	90
13	Alan Scott	Male	blue	-	Blond	180	DC Comics	-	good	90
14	Alex Mercer	Male	-	Human	-	-99	Wildstorm	-	bad	-99
15	Alex Woolsly	Male	-	-	-	-99	NBC - Heroes	-	good	-99
16	Alfred Pennyworth	Male	blue	Human	Black	178	DC Comics	-	good	72
17	Alien	Male	-	Xenomorph XX121	No Hair	244	Dark Horse Comics	black	bad	169
18	Allan Quatermain	Male	-	-	-	-99	Wildstorm	-	good	-99
19	Amazo	Male	red	Android	-	257	DC Comics	-	bad	173

only showing top 20 rows

Question 2)

```
df.show(10)
```

Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publisher	Skincolor	Alignment	Weight
0	A-Bomb	Male	yellow	Human	No Hair	203	Marvel Comics	-	good	441
1	Abe Sapien	Male	blue	Ichthyo Sapien	No Hair	191	Dark Horse Comics	blue	good	65
2	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
3	Abomination	Male	green	Human / Radiation	No Hair	203	Marvel Comics	-	bad	441
4	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	-99
5	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
6	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	-99
7	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
8	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
9	Agent Bob	Male	brown	Human	Brown	178	Marvel Comics	-	good	81

only showing top 10 rows

Question 3)

```
df1 = df.filter(df.Gender == 'Male')
df1.show()
df1.count()
```

Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publisher	Skincolor	Alignment	Weight
0	A-Bomb	Male	yellow	Human	No Hair	203	Marvel Comics	-	good	441
1	Abe Sapien	Male	blue	Ichthyo Sapien	No Hair	191	Dark Horse Comics	blue	good	65
2	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
3	Abomination	Male	green	Human / Radiation	No Hair	203	Marvel Comics	-	bad	441
4	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	-99
5	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
6	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	-99
7	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
9	Agent Bob	Male	brown	Human	Brown	178	Marvel Comics	-	good	81
10	Agent Zero	Male	-	-	-	191	Marvel Comics	-	good	104
11	Air-Walker	Male	blue	-	White	188	Marvel Comics	-	bad	108
12	Ajax	Male	brown	Cyborg	Black	193	Marvel Comics	-	bad	90
13	Alan Scott	Male	blue	-	Blond	180	DC Comics	-	good	90
14	Alex Mercer	Male	-	Human	-	-99	Wildstorm	-	bad	-99
15	Alex Woosly	Male	-	-	-	-99	NBC - Heroes	-	good	-99
16	Alfred Pennyworth	Male	blue	Human	Black	178	DC Comics	-	good	72
17	Alien	Male	-	Xenomorph XX121	No Hair	244	Dark Horse Comics	black	bad	169
18	Allan Quatermain	Male	-	-	-	-99	Wildstorm	-	good	-99
19	Amazo	Male	red	Android	-	257	DC Comics	-	bad	173
20	Ammo	Male	brown	Human	Black	188	Marvel Comics	-	bad	101

only showing top 20 rows
505

Question 3

```
df2 = df.filter(df.Gender == 'Female')
df2.show()
df2.count()
```

Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publisher	Skincolor	Alignment	Weight
8	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
24	Angel Dust	Female	yellow	Mutant	Black	165	Marvel Comics	-	good	57
25	Angel Salvadore	Female	brown	-	Black	163	Marvel Comics	-	good	54
26	Angela	Female	-	-	-	-99	Image Comics	-	bad	-99
38	Arachne	Female	blue	Human	Blond	175	Marvel Comics	-	good	63
40	Arclight	Female	violet	-	Purple	173	Marvel Comics	-	bad	57
41	Ardina	Female	white	Alien	Orange	193	Marvel Comics	gold	good	98
43	Ariel	Female	purple	-	Pink	165	Marvel Comics	-	good	59
44	Armor	Female	black	-	Black	163	Marvel Comics	-	good	50
51	Atom Girl	Female	black	-	Black	168	DC Comics	-	good	54
55	Aurora	Female	blue	Mutant	Black	180	Marvel Comics	-	good	63
62	Batgirl	Female	-	-	-	-99	DC Comics	-	good	-99
63	Batgirl	Female	green	Human	Red	170	DC Comics	-	good	57
64	Batgirl III	Female	-	-	-	-99	DC Comics	-	good	-99
65	Batgirl IV	Female	green	Human	Black	165	DC Comics	-	good	52
66	Batgirl V	Female	-	-	-	-99	DC Comics	-	good	-99
67	Batgirl VI	Female	blue	-	Blond	168	DC Comics	-	good	61
72	Batwoman V	Female	green	Human	Red	178	DC Comics	-	good	-99
80	Big Barda	Female	blue	New God	Black	188	DC Comics	-	bad	135
85	Binary	Female	blue	-	Blond	180	Marvel Comics	-	good	54

only showing top 20 rows
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Question 5)

```
df3=df.groupBy("Race").count()
df3.show()
```

	Race	count
	Neyaphem	1
	Symbiote	9
	Kryptonian	7
	Ungaran	1
	Martian	1
	Strontian	1
	Human / Radiation	11

Kakarantharaian	1
Metahuman	2
Dathomirian Zabrak	1
Human / Cosmic	2
Xenomorph XX121	1
Animal	4
Frost Giant	2
Cyborg	11
Inhuman	4
God / Eternal	14
Yoda's species	1
Ichthyo Sapien	1
Rodian	1

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only showing top 20 rows

Question 6)

```
from pyspark.sql.functions import col, when, avg
from pyspark.sql.types import FloatType

# Clean 'Height' column: replace '-99' with None and cast to FloatType
df_cleaned = df.withColumn("Height_cleaned",
                           when(col("Height") == "-99", None)
                           .otherwise(col("Height").cast(FloatType()))

# Find average height by publisher using the cleaned height
result = df_cleaned.groupBy("Publisher").agg(avg("Height_cleaned").alias("average_height"))

# Show result
result.show()
```

Publisher	average_height
Rebellion	188.0
HarperCollins	NULL
J. R. R. Tolkien	279.0
Star Trek	181.5
Marvel Comics	190.51080243087108
Wildstorm	NULL
NULL	142.58333333333334
South Park	NULL
Sony Pictures	179.0
Titan Books	183.0
ABC Studios	NULL
SyFy	NULL
Image Comics	211.0
Universal Studios	NULL
IDW Publishing	NULL
NBC - Heroes	NULL
Hanna-Barbera	NULL
DC Comics	180.90068493150685
Microsoft	213.0
J. K. Rowling	NULL

+-----+

only showing top 20 rows

Simple way

```
from pyspark.sql.functions import avg
from pyspark.sql.functions import col

df = df.withColumn("Height", col("Height").cast("float"))
df.groupBy("Publisher").avg("Height").show()
```

Publisher	avg(Height)
Rebellion	188.0
HarperCollins	-99.0
J. R. R. Tolkien	279.0
Star Trek	-5.5
Marvel Comics	142.75644326701607
Wildstorm	-99.0
NULL	-2.3666666666666667
South Park	-99.0
Sony Pictures	179.0
Titan Books	183.0
ABC Studios	-99.0
SyFy	-99.0
Image Comics	-76.85714285714286
Universal Studios	-99.0

IDW Publishing	-99.0
NBC - Heroes	-99.0
Hanna-Barbera	-99.0
DC Comics	91.07209302325582
Microsoft	213.0
J. K. Rowling	-99.0

only showing top 20 rows

Question 7)

```
df.filter(df.Eyecolor == 'blue').show()
```

Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publisher	Skincolor	Alignment	Weight
1	Abe Sapien	Male	blue	Ichthyo Sapien	No Hair	191	Dark Horse Comics	blue	good	65
2	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
4	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	-99
5	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
6	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	-99
7	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
8	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
11	Air-Walker	Male	blue	-	White	188	Marvel Comics	-	bad	108
13	Alan Scott	Male	blue	-	Blond	180	DC Comics	-	good	90
16	Alfred Pennyworth	Male	blue	Human	Black	178	DC Comics	-	good	72
22	Angel	Male	blue	-	Blond	183	Marvel Comics	-	good	68
27	Animal Man	Male	blue	Human	Blond	183	DC Comics	-	good	83
29	Ant-Man	Male	blue	Human	Blond	211	Marvel Comics	-	good	122
30	Ant-Man II	Male	blue	Human	Blond	183	Marvel Comics	-	good	86
33	Anti-Venom	Male	blue	Symbiote	Blond	229	Marvel Comics	-	-	358
35	Aquababy	Male	blue	-	Blond	-99	DC Comics	-	good	-99
36	Aqualad	Male	blue	Atlantean	Black	178	DC Comics	-	good	106
37	Aquaman	Male	blue	Atlantean	Blond	185	DC Comics	-	good	146
38	Arachne	Female	blue	Human	Blond	175	Marvel Comics	-	good	63
39	Archangel	Male	blue	Mutant	Blond	183	Marvel Comics	blue	good	68

only showing top 20 rows

Start coding or [generate](#) with AI.